



CCSW-225 Human Computer Interaction

2025 2nd semester

Group Project

Project Title:	SeedSync App	
Assignment#	Final LAB report	
Submission Date:	2025, April 22	
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Section:	SE3, Group3	

- Contribution Log -

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no.	Task	M1	M2	M3	M4	Notes
1	Project Overview			-		
2	Targeted Users		-			
3	Main Problem	-				
4	Collecting Data and Find Users		-			
5	Interview Questions	-	-	-	-	
6	Online Questioner Questions	-	-	-	-	
7	Create the online Questions				-	
8	Interview 1 Summaries				-	
9	Interview 2 Summaries			-		
10	Interview 3 Summaries	-				
11	Interview 4 Summaries		-			
12	survey overview (Visualization)	-				

13	Need and Insight 1				-	
14	Need and Insight 2				-	
15	Need and Insight 3			-		
16	Need and Insight 4	-				
17	Need and Insight 5		-			
18	Tasks	-	-			
19	Excel sheet				-	
20	Motor Control Pages Low-fidelity			-		
21	Sensor Reading Pages Low-fidelity	-				
22	Choose Greenhouse Pages Low-fidelity	-				
23	Add greenhouse Pages Low-fidelity		-			
24	How the design work				-	
25	How does the design meet user needs				-	
26	Splash Screen		-			
27	Motor Control Pages High-fidelity			-		
28	Sensor Reading Pages High-fidelity	-				
29	Choose Greenhouse Pages High-fidelity	-		-		
30	Add greenhouse Pages High-fidelity		-		-	

31	Additional Pages Added for Functional Completion	-	-	-		
32	Short Explanation of the Application/System (with HCI principles)		-			
33	The system applies core HCI principles through its design features		-			
34	Review	-		-		
35	Edit	-				

- Deliverables Log -

	Mark	Due Date	Instructor Notes
Assignment#1			
General project idea.		Week 3	
Targeted users.			
Main problems to be solved.			
How to: Data collection.	1		
How to: find users.	1		
Interview questions.	3		
Online questioner link.	3		
Assignment#2			
The raw data collected from the users.		Week 4	
Visualization of collected data.	2.5		
List of 5 needs.	1.5		
List of 5 insights.	1.5		
List of 5 main tasks.	1.5		
Assignment#3			
Photos of Low-Fidelity Prototyping for 3 distinct tasks	4	Week 5	
1-2 sentence caption for each significant screen.			
How will the new design meet users' requirements?			

Assignment#4			
High-Fidelity Interactive Prototype for 3 distinct tasks	6	Week 7	
Explanation of all interfaces			
Explanation of the designed application or system			
Assignment#5			
Final project report.		Week 7	
The presentation file.	5	Week 10	
Total	30		

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Project Overview

Project Name

SeedSync

Project Idea

The SeedSync app is an innovative solution to aid users efficiently using IoT (Internet of Things) technology. It is designed to automate and optimize greenhouse environment management. IoT plays an important role in monitoring critical environmental parameters such as temperature, humidity, soil moisture and light intensity in real time. In addition, it allows users to remotely control actuators such as fans, water pumps, windows and lights through a user-friendly mobile interface. In the case of no internet connection in the user's device, the system will automatically save data in the database and display the past data to the user when they are back online. The aim of SeedSync is to enhance crop yield, reduce manual labor and ensure optimal growing for plants.

Workflow

1. Sensors collect data and send it to the microcontroller.
2. The microcontroller sends data to the cloud via Wi-Fi.
3. The backend processes the data and stores it in a database.
4. The app gets data from the backend and displays it to the user.
5. The user sends control commands through the app, which are transmitted to the microcontroller to control actuators.

Targeted Users

1. Greenhouse Business Owners: people who own or operate greenhouses, often for commercial or large-scale growing purposes.
2. Farm Managers: Technicians responsible for overseeing and maintaining multiple greenhouses, often on behalf of a farm or business.
3. Home Gardeners/ Hobbyists: individuals who grow plants for personal enjoyment, food, or decoration often in small-scale or urban settings.

Main Problem

Owners of greenhouses or home gardens face many problems, the most important of which is effectively managing environmental conditions such as temperature control, humidity management and water supply. Addressing these problems requires careful planning, continuous monitoring and investment in technology. The SeedSync application aims to optimize monitoring greenhouses and controlling its factors in a more efficient and less time-consuming way.

Collecting Data and Finding Users

1. Conducting one-on-one interviews with individuals from our target groups from the team members connections.
2. Publishing a survey through social media platforms where groups of similar interests often gather and interact such as reddit, and X (formerly Twitter) communities.

Interview Questions

1. Do you have a greenhouse or a garden?
2. When did you start it?
3. What type of greenhouse do you have? Is it for personal use, small-scale farming, or something else?
4. How do you manage temperature, humidity, and lighting in your greenhouse?
5. How much time do you typically spend managing your greenhouse or garden each week?
6. What tasks do you find most time consuming?
7. What challenges did you face while taking care of your greenhouse or garden?
8. Do you have experience with automated monitoring?
9. (If yes) are they effective?
10. (If not) Would you be open to the idea?

Online Questioner

<https://forms.office.com/Pages/ResponsePage.aspx?id=G9ITtMFqYUu4nFSISCljP3zHedNUCNZOvFpyglF9ritUO1dXMjM0VTBPTDMwR1hDNVlKUfXNFdRVS4u>

Questioner questions:

1. Do you currently have a greenhouse?
 - ☐ Yes
 - ☐ No
2. What's stopping you?
 - ☐ Financial difficulties
 - ☐ Don't know how to take care of them
 - ☐ Time consuming
 - ☐ There's no space in my current home
 - ☐ Others
3. How many greenhouses do you currently manage?
 - ☐ **None**
 - ☐ **1**
 - ☐ **2–5**
 - ☐ **6–10**
 - ☐ **More than 10**
4. How large is your greenhouse?
 - ☐ **Small** (less than 10 square meters)
 - ☐ **Medium** (10–50 square meters)
 - ☐ **Large** (50–100 square meters)
 - ☐ **Very Large** (over 100 square meters)
 - ☐ **I don't have a greenhouse**

5. How often do you experience issues with your greenhouse due to environmental factors?
 - ☐ Always (constantly facing issues)
 - ☐ Frequently (monthly or more often)
 - ☐ Occasionally (a few times a year)
 - ☐ Rarely (once or twice a year)
 - ☐ Never
6. Do you use any systems to monitor environmental conditions in your greenhouse?
 - ☐ Yes
 - ☐ No
 - ☐ I do not have a greenhouse
7. What are the main challenges you face in managing your greenhouse? (You can select more than one)
 - ☐ Monitoring temperature
 - ☐ Controlling humidity
 - ☐ Managing irrigation
 - ☐ Ensuring proper lighting
 - ☐ Other
8. What features would you find useful in a greenhouse monitoring system? (You can select more than one)
 - ☐ Real-time monitoring of temperature, humidity, and light
 - ☐ Alerts on your phone
 - ☐ Automatic irrigation system
 - ☐ Data analysis to improve productivity
 - ☐ Other
9. How would you prefer to view data in the app?
 - ☐ Graphs and charts
 - ☐ Numerical values
 - ☐ Color-coded indicators (green = good, red = bad)
 - ☐ A combination of the above
10. Would you be interested in using our mobile application?
 - ☐ Very likely
 - ☐ Somewhat likely
 - ☐ Neutral
 - ☐ Somewhat unlikely
 - ☐ Very likely
 - ☐
11. Suggestions? (Not mandatory)

Interview Summaries:

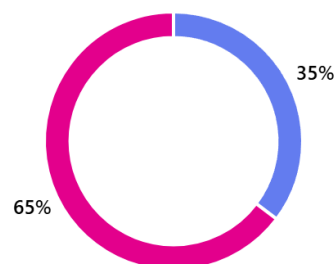
Interview No.	Interviewer	Interviewee	Interview Content
1	Suha Algarni	The interviewee's grandfather	The Interviewee owns a large garden with palm trees and seasonal vegetables but does not have a greenhouse. He faces challenges such as fluctuating weather, water scarcity, and pests. Tasks like watering and weeding take 10 to 15 hours per week, which is difficult due to his back problems. He hasn't used modern monitoring or automated irrigation systems and relies on traditional methods. After the interview, he learned about greenhouses and systems that monitor soil moisture and automate irrigation. He is now considering building a greenhouse and showed interest in using easy, effective tools to reduce physical effort and improve garden management.
2	Reman Alamoudi	A greenhouse owner in California	The interviewee has 5 years of gardening experience and has owned a personal-use greenhouse for 1 year. Their greenhouse, located in California, faces challenges with extreme heat and humidity. They currently use shade cloths, manual spraying, and a Bluetooth-enabled hygrometer for monitoring, but struggle with app usability and limited Bluetooth range. Tasks like watering and adjusting ventilation are time-consuming, and they face issues with winter moisture and insufficient ventilation. They are interested in automation (e.g., motorized windows, fans) and remote monitoring but emphasize the need for affordability, ease of use, and reliable functionality. They suggested a tiered pricing model and a user-friendly UI for the proposed IoT app to cater to both hobbyists and larger operations.
3	Marwa Hadaddi	Rawan Adnan, specialized in rooftop, terrarium and bonsai cultivation	The interviewee has a large home garden for personal and commercial use and started growing it 12 years ago. She mentioned that the main problems she faces are managing heat and humidity. She mentioned that managing and monitoring the soil is what takes the most time. The biggest challenge she faces is the time it takes to carry out some daily agricultural tasks. She also mentioned that she used a temperature monitor but had to use manual processes to deal with humidity and lighting levels, which were effective but took a lot of time and effort.
4	Shumokh Alsharif	A local farmer from reddit greenhouse communities	The interviewee owns three medium-sized greenhouses used for small-scale commercial farming, growing vegetables for local markets. He started 4 years ago and manage his greenhouse with shade nets, fans, and analog hygrometers. Although he has an automated irrigation system, it lacks precision in humidity control, which has led to fungal diseases and pest outbreaks. He also struggles with condensation dripping from the roof onto the crops. After the interview he expressed interest in automated monitoring especially if it includes pest and humidity management features. He also suggested that the system should be

			adaptable to different greenhouse sizes and budgets allowing small farmers like him to use it without a high cost.
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Visualization:

2. Do you currently have a greenhouse? هل لديك دفيئة زراعية حالياً؟

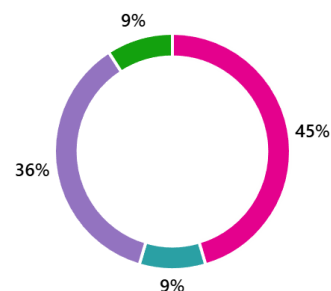
- Yes , نعم 6
- No , لا 11



This question assesses the ownership status of greenhouses among respondents. A majority do not own a greenhouse, indicating a potential market for educational resources and support for potential greenhouse owners.

3. What's stopping you?

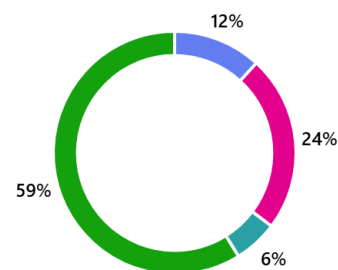
- Financial difficulties , صعوبات مالية 0
- Don't know how to take care of them , لا أعرف كيفية العناية بها 5
- Time consuming , تستهلك الوقت 1
- There's no space in my current home , لا يوجد مكان في منزلي الحالي 4
- Other 1



This question identifies barriers to greenhouse ownership. The most significant obstacle is a lack of knowledge about greenhouse care, followed by space limitations. Financial difficulties are not a concern for any respondents.

4. How many greenhouses do you currently manage?

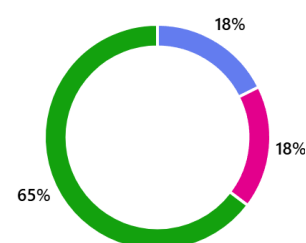
- 1 2
- 2 - 5 4
- 6 - 10 1
- More than 10 , أكثر من 10 0
- None , لا ادير حالياً 10



This question measures the number of greenhouses respondents have. The majority do not operate any greenhouses, while 12% operate one and 24% operate 2-5 greenhouses. This highlights the need for entry-level features.

5. How large is your greenhouse?

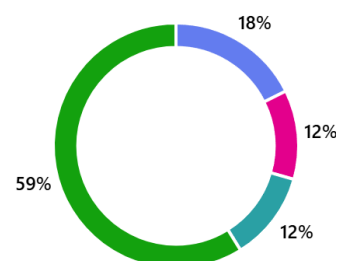
- Small (less than 10 square meters) , صغيرة (أقل من 10 متر مربع) 3
- Medium (10–50 square meters) , متوسطة (10–50 متر مربع) 3
- Large (50–100 square meters) , كبيرة (50–100 متر مربع) 0
- Very Large (over 100 square meters) , كبيرة جداً (أكثر من 100 متر مربع) 0
- I don't have a greenhouse , ليس لدي دفيئة 11



This question examines the size of existing greenhouses among respondents. Only 18% own small or medium-sized greenhouses, while 65% do not have one at all.

6. How often do you experience issues with your greenhouse due to environmental factors?

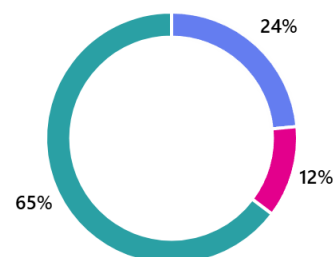
- Always (constantly facing issues) , دائماً (أواجه مشاكل بشكل مستمر) 3
- Frequently (monthly or more often) , بشكل متكرر (شهرياً أو أكثر) 2
- Occasionally (a few times a year) , أحياناً (عدة مرات في السنة) 2
- Rarely (once or twice a year) , نادراً (مرة أو مرتين في السنة) 0
- Never , أبداً 10



This question explores how frequently respondents face environmental issues in their greenhouses. Around 40% of respondents stated they faced issues while managing their greenhouse, suggesting many users may benefit from preventive guidance.

7. Do you use any systems to monitor environmental conditions in your greenhouse?

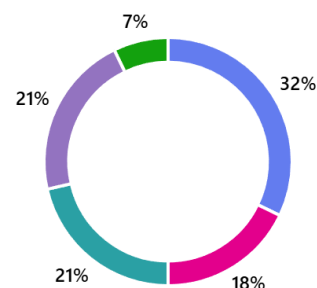
- Yes , نعم 4
- No , لا 2
- I do not have a greenhouse, ليس لدي دفيئة 11



This question assesses the use of monitoring systems. Only 24% of respondents utilize such systems, indicating an opportunity for the app to provide valuable monitoring features.

8. What are the main challenges you face in managing your greenhouse? (You can select more than one)

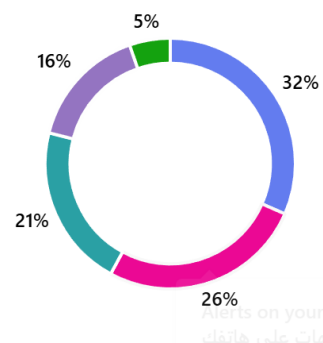
- Monitoring temperature , مراقبة درجة الحرارة 9
- Controlling humidity , التحكم في الرطوبة 5
- Managing irrigation , إدارة الري 6
- Ensuring proper lighting , ضمان الإضاءة المناسبة 6
- Other 2



This question identifies key management challenges. Important issues include monitoring temperature (32%) and managing irrigation (21%). The app could incorporate tools to help address these specific challenges.

9. What features would you find useful in a greenhouse monitoring system? (You can select more than one)

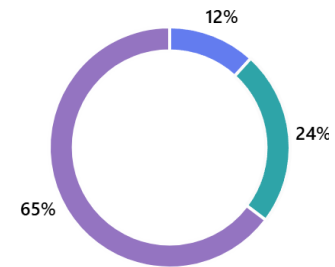
- Real-time monitoring of temperature, humidity, and light , المراقبة الفورية لدرجة الحرارة، الرطوبة، والإضاءة 12
- Alerts on your phone , التنبيهات على هاتفك 10
- Automatic irrigation system , نظام ري تلقائي 8
- Data analysis to improve productivity , تحليل البيانات لتحسين الإنتاجية 6
- Other 2



This question seeks to understand desired features for monitoring systems. Respondents expressed interest in real-time monitoring (32%) and alerts (26%), highlighting important functionalities for the app.

10. How would you prefer to view data in the app?

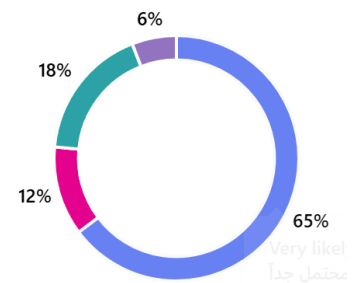
- Graphs and charts , الرسوم البيانية والمخططات 2
- Numerical values , القيم الرقمية 0
- Color-coded indicators (green = good, red = bad) , مؤشرات ملونة (أخضر = جيد، أحمر = سيء) 4
- A combination of the above , مزيج من الخيارات السابقة 11



This question addresses data display preferences. Most participants prefer a combination of display methods, indicating that the application should provide multiple options for displaying information.

11. Would you be interested in using our mobile application?

- Very likely , من المحتمل جداً 11
- Somewhat likely , من المحتمل إلى حد ما 2
- Neutral , محايد 3
- Somewhat unlikely , من غير المحتمل إلى حد ما 1
- Very unlikely , من غير المحتمل جداً 0



This question assesses overall interest in the app. A strong 65% of respondents indicated they would be very likely to use it, indicating positive potential for user adoption and engagement.

12. Suggestions?

4 Responses

ID ↑	Name	Responses
1	anonymous	I need a device to measure temperature, take actions based on temp, have a physical display and also an app with data logging
2	anonymous	متحمسة جداً لتجربة تطبيقكم المحمول
3	anonymous	ش
4	anonymous	a lot of greenhouses manage their temperature and humidity by opening vents and running vents, and the technology for the vents to open at a certain temperature is very simple and cheap, so just keep in mind that you need to offer people something better than that for those on a budget. Good luck!

This question allowed respondents to provide open-ended feedback. Responses varied, offering insights into desired features and improvements, emphasizing the need for practical tools and user-friendly technology. This qualitative data can help shape the app's development to better meet user needs.

Needs, Insights:

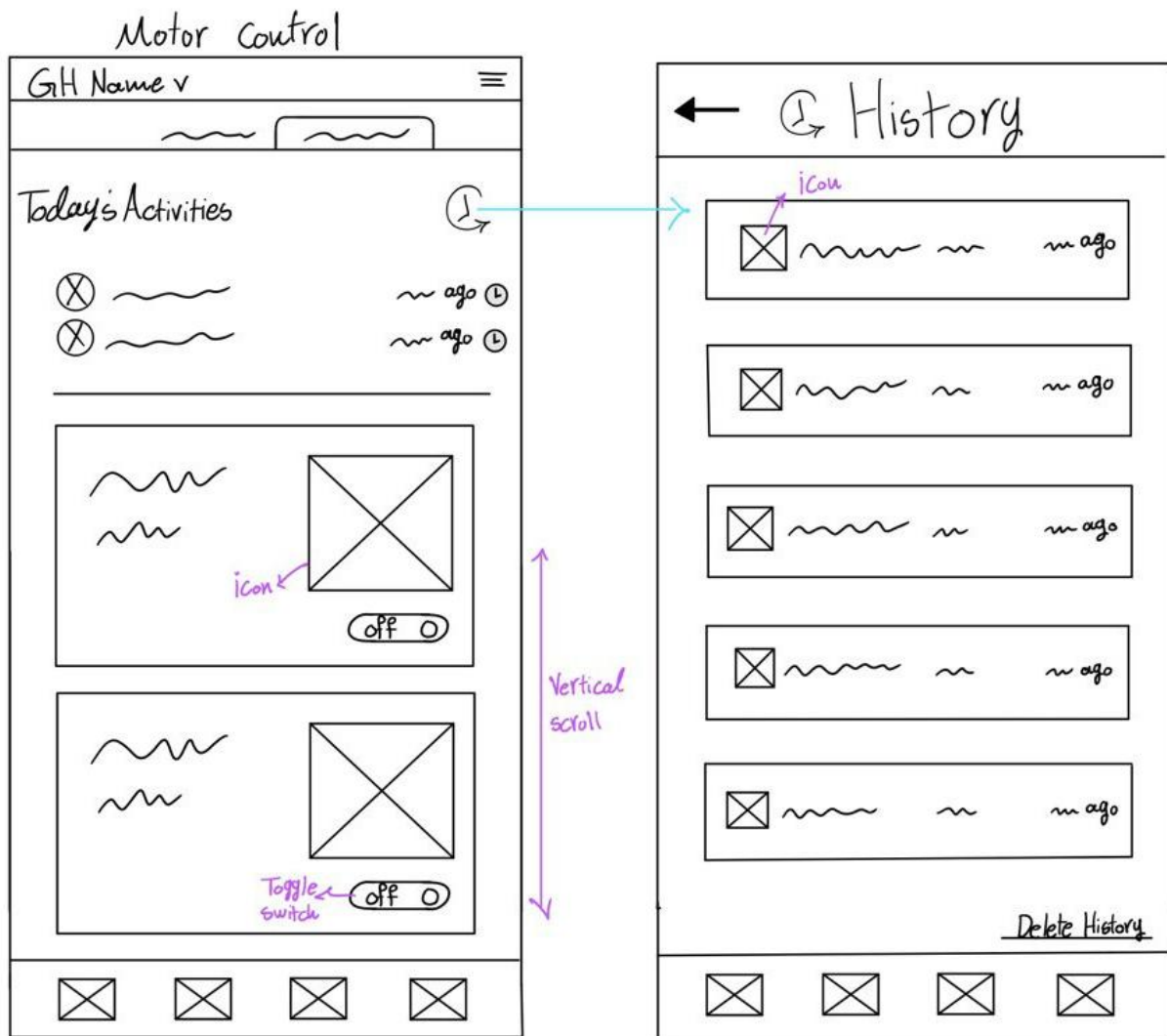
No	User	Needs	Insights
1	The interviewee's grandfather	An easier process of watering plants to reduce physical effort.	The user experience weakness due to old age. He feels that the burden of watering plants daily to be an exhausting task and would prefer it if it could be replaced by other means.
2	The interviewee's grandfather	A way to monitor soil moisture levels accurately without manual checking.	The user seems to have difficulty with monitoring soil moisture levels manually. This is probably the case because manual measuring of moisture in soil requires kneeling down to put your finger in the soil. This causes increased back pain.
3	A greenhouse owner in California	A tiered pricing model	Users require a flexible pricing model that aligns with their specific needs, as the system will be used across a wide range of contexts—from large-scale agricultural greenhouses to personal home gardens. A one-size-fits-all pricing approach could exclude certain user groups, particularly individuals with limited budgets or smaller-scale needs. A tiered pricing structure would ensure accessibility and inclusivity, making the system viable for both commercial and personal users while enhancing overall user satisfaction.
4	Rawan Adnan, specialized in rooftop, terrarium and bonsai cultivation	To accurately monitor pH levels in soil	The user doesn't have a way to know the changes in the pH level in the soil by looking or touching, and using pH strips takes time and effort and does not monitor them periodically. The user prefers to be able to monitor the pH in the soil periodically and quickly to take the necessary measures at the appropriate time.
5	A local farmer from reddit greenhouse communities	To prevent condensation buildup on the greenhouse roof during winter	The user struggles with condensation forming on the greenhouse roof during winter. As temperatures drop at night, warm humid air inside the greenhouse condenses on the cold roof panels, leading to water droplets falling onto crops, increasing the risk of fungal diseases.

Tasks:

- **Remote monitoring:** Implement IoT technology to allow users to access the app from anywhere with an internet connection, and present data in an intuitive and user-friendly UI, displaying:
 - Humidity
 - Soil moisture
 - Temperature
 - Carbon dioxide levels
 - Soil pH
 - Wind speed and direction
 - Precipitation levels
 - Light intensity
- **Motor control:** Enable the following automated control of greenhouse motors:
 - **Ventilation Motors (Fans)**
 - **Shade Motors (Shade Cloth/Curtain Systems)**
 - **Irrigation Motors (Water Pumps)**
 - **Window/Door Actuators**
 - **Heating/Cooling Motors (HVAC Systems)**
 - **Misting Motors (Humidity Control)**
- **Automated actions:** Implement system responses based on sensor readings.
- **Offer 3 service levels to accommodate different budgets and needs:**
 - Personal use: affordable option with limited functionalities for hobbyists or small setups.
 - Small scale garden: moderately priced with enhanced functionalities for home gardens or small farms.
 - Large scale agriculture farming: full-featured, high-capacity solution for larger farming operations.
- **Multi-greenhouse management:** support the monitoring and control of multiple greenhouses within a single interface.
- **Greenhouse specification:** allow users to input specific greenhouse details for optimized performance and accurate reading analysis.

Low-Fidelity sketches

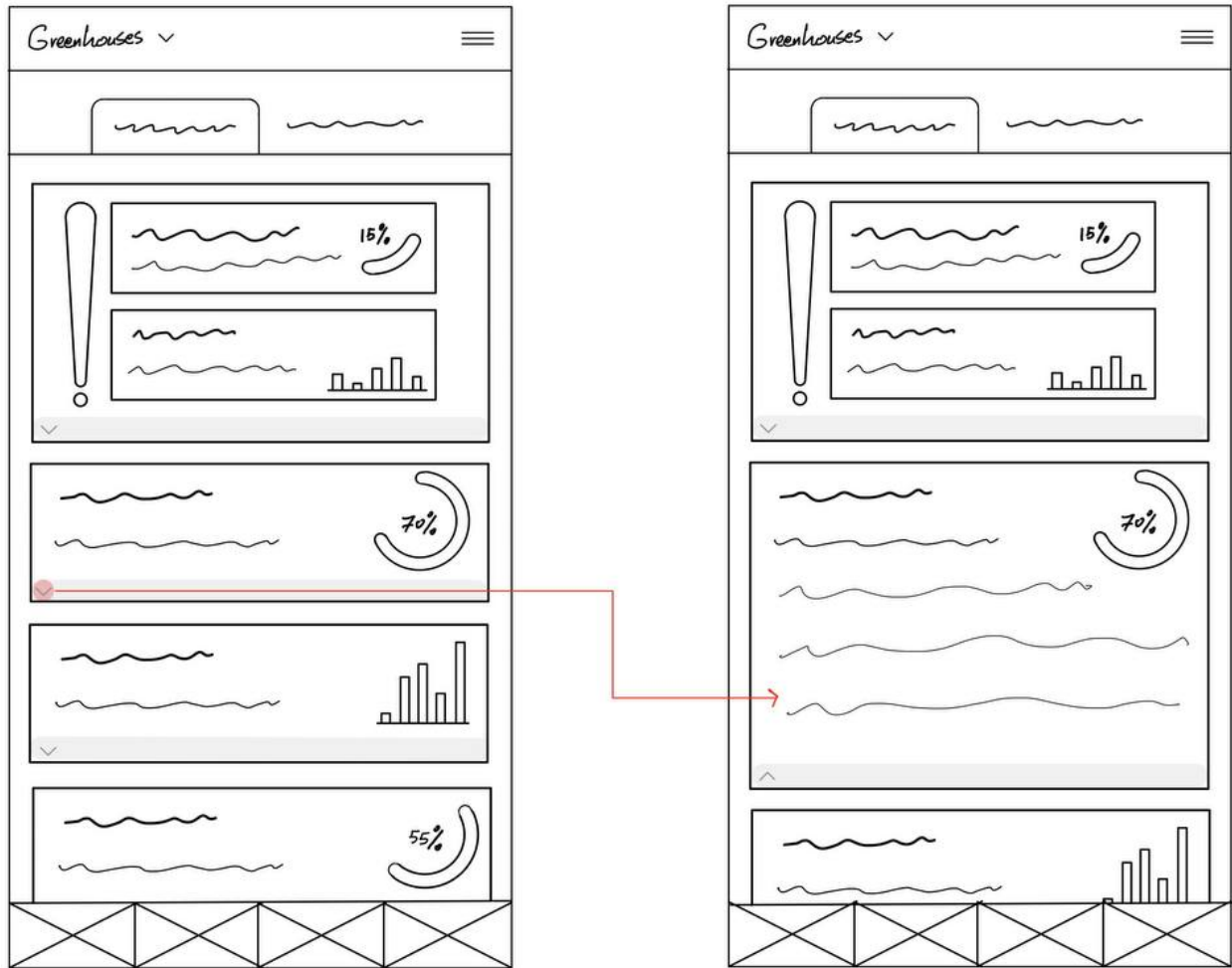
Motor Control Pages:



The motor control page displays the most recent activities at the top, and cards that shows the motors status with the option to display more info.

The history page lists all previous activities with date stamps along with an icon that clarify if the action was manual or automated.

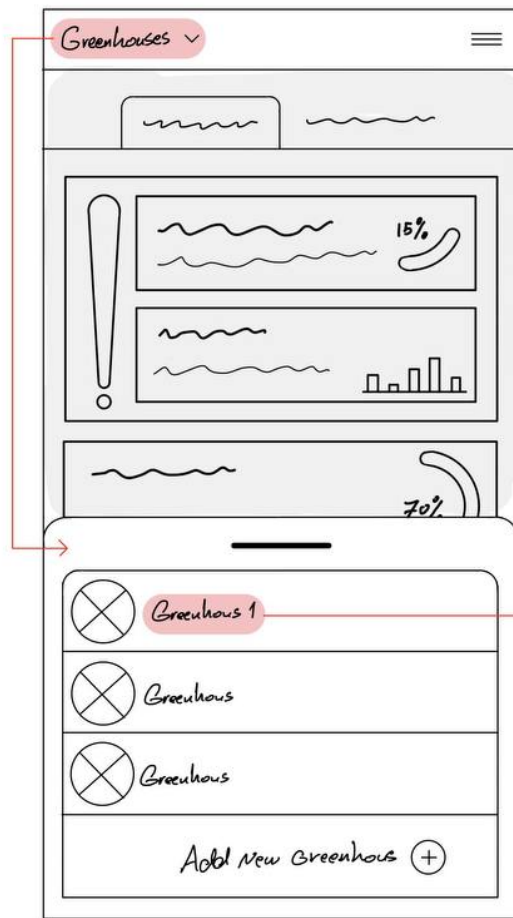
Sensor Reading Pages:



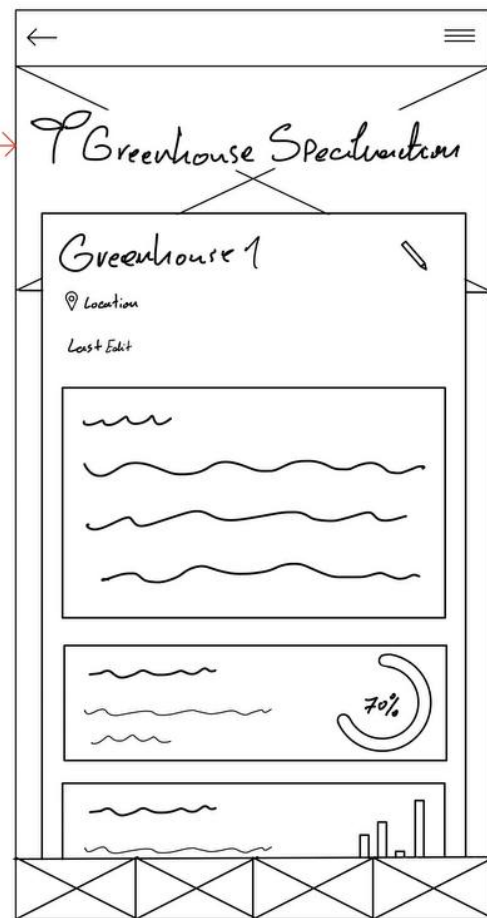
Readings page: a Vertical scroll Displays readings of monitoring the Greenhouse. At the beginning of these readings, we see the dangerous alarms box and after that, the readings are arranged from the most dangerous to the least dangerous.

We see a small gray arrow below each box. If you click on this arrow, more details about the readings for this box will be displayed.

Choose Greenhouse Pages:



When you click on the arrow next to the word Greenhouse at the top of the page, all the Greenhouses that you manage will be displayed.



When you click on any one of them, the Greenhouse Specification page will be displayed.

This page displays an accurate description of the Greenhouse, a picture of it, and also its readings. Through the icon next to the Greenhouse name, the user can edit the data for his Greenhouse.

Add Greenhouse Pages:

The image displays three sequential wireframe screens for an application titled "Add greenhouse".

- Screen 1:** Features a back arrow, a title "Add greenhouse", and a "Cancel" button. Below is a progress bar with five segments, the first of which is filled. The main content area is titled "Type of Greenhouse" and contains three selectable options, each with a placeholder icon and a text field for a description:
 - Personal description
 - Small description
 - Agri descriptionA "Next" button is at the bottom.
- Screen 2:** Features a back arrow, a title "Add greenhouse", and a "Cancel" button. Below is a progress bar with five segments, the second of which is filled. The main content area is titled "Size" and contains a large square placeholder for a dynamic animation. Below this are three input fields labeled "Width", "Height", and "Length". A "Next" button is at the bottom.
- Screen 3:** Features a back arrow, a title "Add greenhouse", and a "Cancel" button. Below is a progress bar with five segments, the third of which is filled. The main content area is titled "Sun Orientation" and contains a square placeholder for a dynamic animation. Below this is a compass rose with labels for North, South, East, and West. A "Next" button is at the bottom.

Blue arrows indicate the flow from the "Next" button of Screen 1 to Screen 2, and from the "Next" button of Screen 2 to Screen 3.

A progress bar informs the user of their progress.

A dynamic animation will be displayed when the user enters their greenhouse size. The animation will change size depending on the input.

The user is asked to choose the sun orientation according to their greenhouse from the compass which will cause the animation to dynamically change to meet the user's input. (The next button will move the user to the next page)

Hand-drawn wireframe of a mobile app for adding a greenhouse. The interface consists of three panels:

- Panel 1 (Add greenhous):** Features a back arrow, a "Cancel" button, a large placeholder for a photo, a text input for "Add name?" (placeholder: "Name..."), three photo selection icons labeled "icon", "gallery", and "camera", and "Skip" and "Next" buttons at the bottom.
- Panel 2 (Add greenhous):** Features a back arrow, a "Cancel" button, a "Location" label, a text input with a location pin icon, and a "Save" button at the bottom.
- Panel 3 (Confirmation):** Features a large placeholder for a confirmation message, the text "Greenhouse saved!", "Summary info" (represented by wavy lines), a "Homepage" button, and a note "goes to the main page" with an arrow pointing to the "Homepage" button.

Blue arrows indicate the flow from the first panel to the second, and from the second panel to the third.

An optional name and photo can be assigned to the greenhouse which the user can skip.

The user is asked to insert their location in order to give more specific information for their greenhouse. The user can either type their location or have it assigned automatically by the GPS.

After the user entered all the details, a confirmation page appears to the user to inform them that their greenhouse has been usefully added. Summary info is displayed on the confirmation page

The design focuses on a smooth and interactive user experience, making greenhouse management easy and clear, even for those without technical expertise.

How It Works?

❖ Motor Control Page

Upon opening the application, the user is presented with the latest activities at the top of the motor control page, along with cards displaying motor status and an option to expand for more details. This ensures quick access to essential greenhouse operations without unnecessary complexity.

❖ History Page

For users who need to review past activities, the history page provides a chronological list of all operations, with icons indicating whether each action was manual or automated. This feature allows users to track system behaviour and monitor past changes efficiently.

❖ Progress Bar & Interactive Visualization

During Greenhouse Setup, a progress bar guides users through the process, keeping them informed about their current step. Simultaneously, a dynamic animation updates automatically based on the entered greenhouse size, providing a clear visual representation to make setup more intuitive.

❖ Adding a Name, Photo, and Location

Users can personalize their greenhouse by assigning a name and photo, or they may skip this step. Additionally, users can enter their location manually or allow GPS to detect it automatically, ensuring precise data and location-based recommendations.

❖ Selecting Sun Orientation & Greenhouse Size

The system prompts users to set the sun orientation using a compass interface. As they adjust the direction, the animation dynamically updates, helping them optimize greenhouse positioning for maximum sunlight exposure.

❖ Confirmation Page

Before finalizing the setup, a confirmation page appears, summarizing all entered information, allowing users to review and verify their inputs before saving.

❖ Readings Page

When users access the readings page, a critical alerts box is prominently displayed at the top to ensure urgent warnings are immediately noticed. Other readings are listed from most critical to least, and users can expand each section for additional details by clicking on the small grey arrow.

❖ Managing Multiple Greenhouses

For users managing multiple greenhouses, a dropdown menu allows seamless switching between them. Selecting a greenhouse opens its specifications page, where users can view detailed descriptions, images, and live readings. The edit icon allows users to modify greenhouse data when needed.

How Does This Design Meet User Needs?

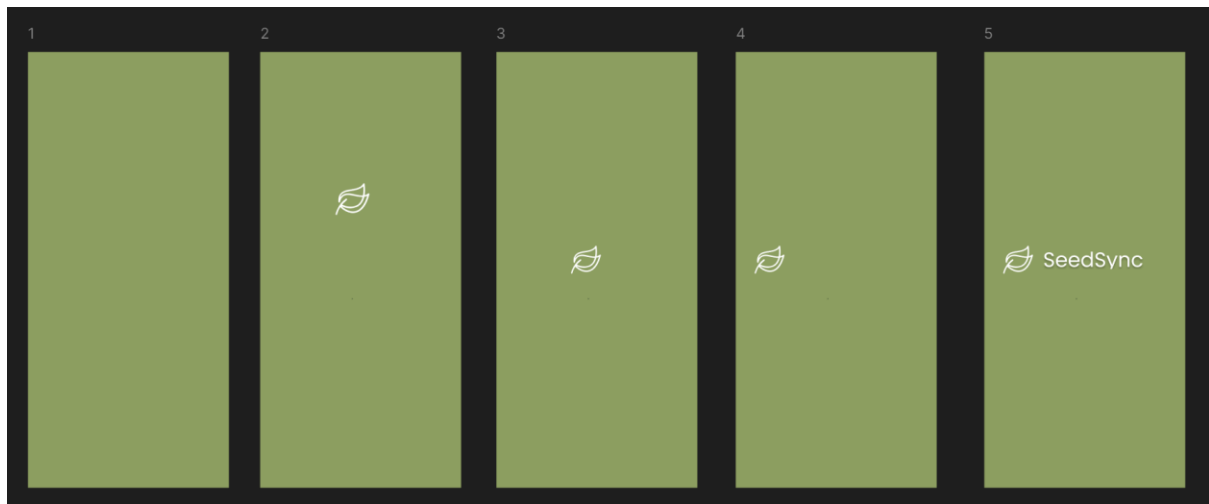
- ❖ **Easy Monitoring & Management**
The interface presents clear, structured information, enabling users to assess their greenhouse status quickly without unnecessary navigation.
- ❖ **Better Decision-Making**
Interactive visual elements, such as dynamic animations and sun orientation visualization, help users optimize their greenhouse setup based on real-time feedback.
- ❖ **Flexible & Personalized Experience**
Users can customize their greenhouse details and choose how to enter their location, making the system adaptable to different user needs.
- ❖ **Increased Efficiency**
By prioritizing important alerts and organizing data from most critical to least, the system ensures users focus on what truly matters without being overwhelmed.
- ❖ **Error Prevention**
The confirmation page helps users verify their inputs before saving, reducing the risk of mistakes.
- ❖ **Seamless Multi-Greenhouse Management**
Users with multiple greenhouses can easily switch and manage them from a single interface, improving workflow and accessibility.

High fidelity Prototype

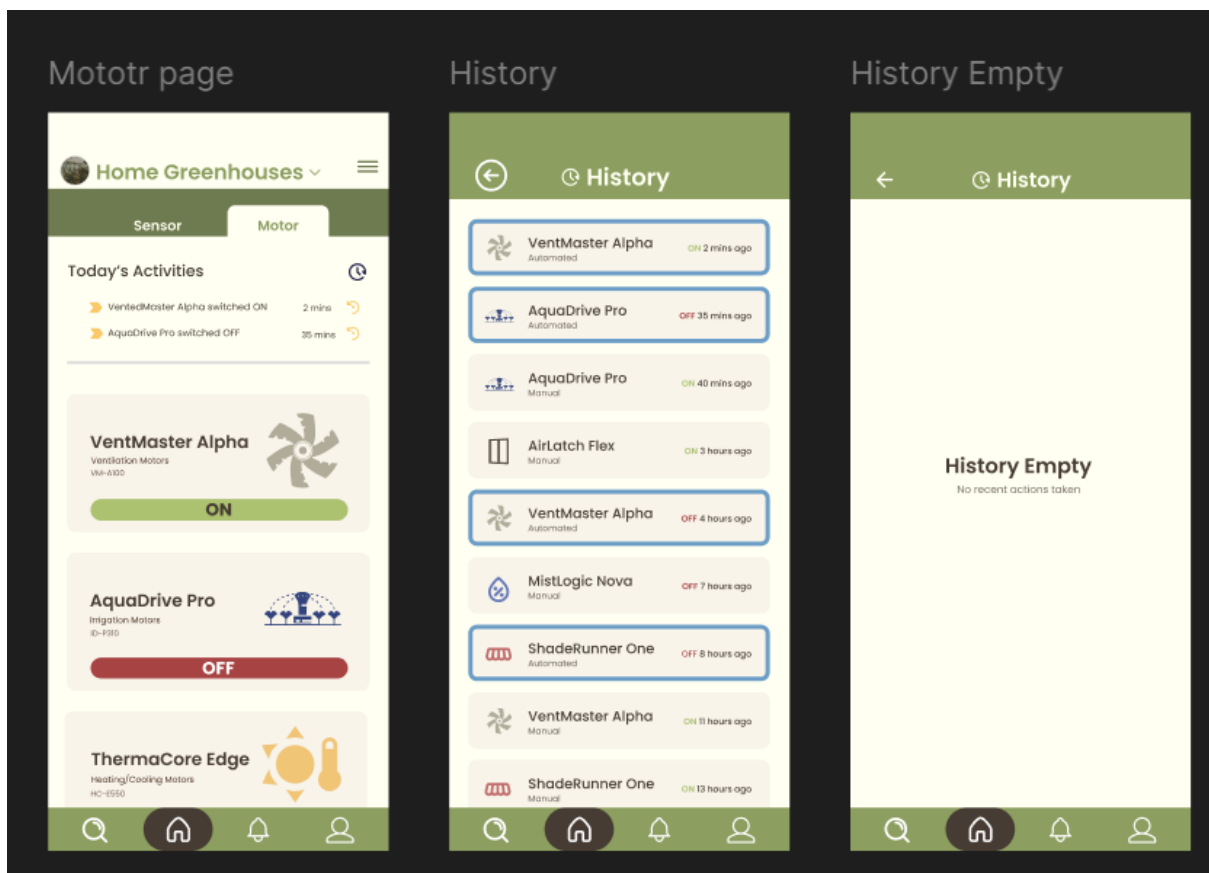
Prototype Link:

<https://www.figma.com/proto/coRQ80n2HJ1Midos33r95y/SeedSync?page-id=0%3A1&node-id=169-400&p=f&viewport=949%2C78%2C0.18&t=HiakqBwhfTBYKnev-1&scaling=scale-down&content-scaling=fixed&starting-point-node-id=345%3A842&show-proto-sidebar=1>

Animated Splash Screen



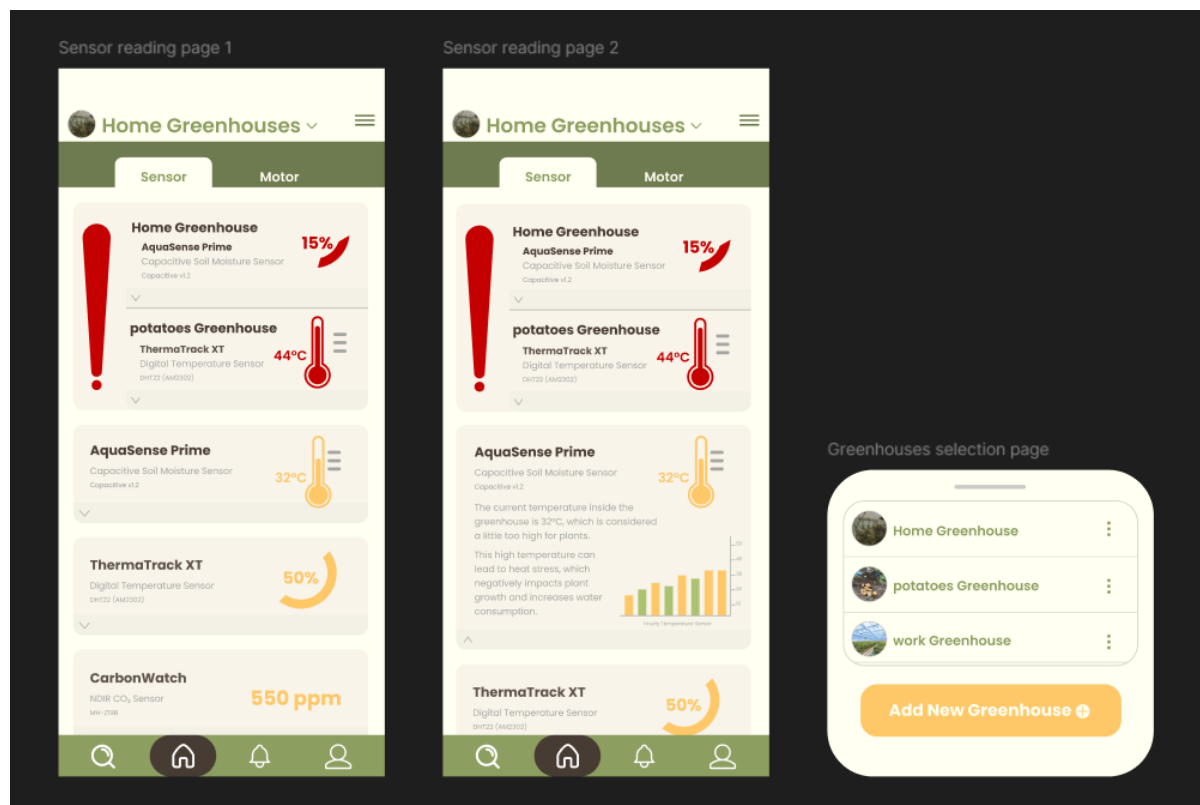
Motor Control Pages



This screen allows users to easily control their greenhouse motors. It displays the automated tasks that the app carried out while the user was away at the top of the page in “Today’s Activities”. The icon beside it is clickable and takes the user to the history when clicked on it. Below are multiple cards containing the most important cards of each motor that the user can turn on and off whenever they wish too.

The history of motors displays all the actions taken by the app whether it was manually via the users order or automatically carried out by the software’s analysis of current reading. At the bottom of the page there is a delete history button in which the user can delete all the past actions if they deem it unnecessary.

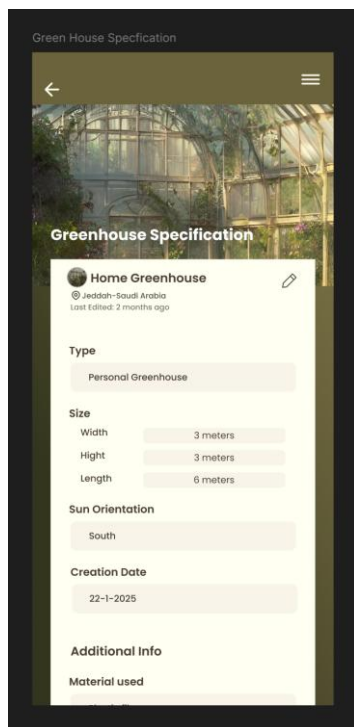
Sensors Readings Pages (Home Page):



The first page shows the critical sensor readings for all Greenhouses at the top, followed by readings for Greenhouses selected from the top of the page, arranged from most critical to least critical. When you click on the gray arrow below each reading box, more details for this sensor reading will appear

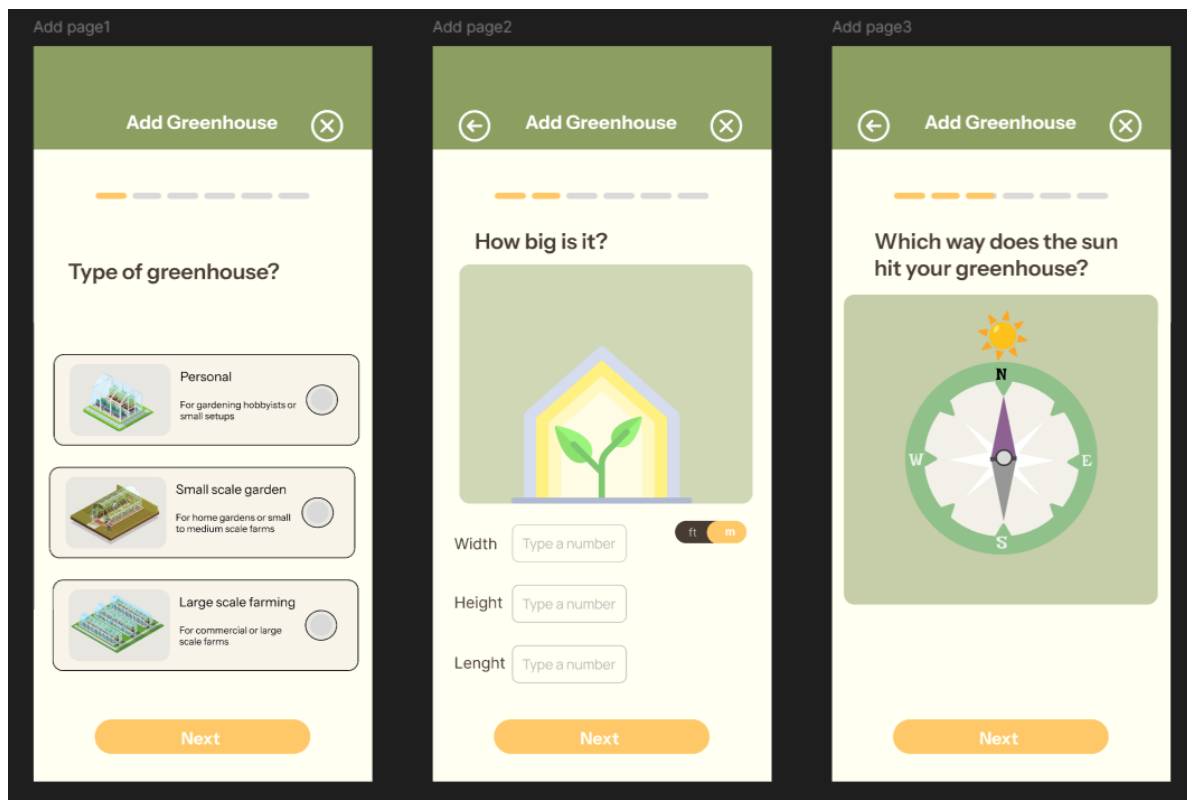
The third page appears when the user clicks on the greenhouse name at the top of the home page. It displays all the greenhouses the user has added. Clicking on the greenhouse name on this page displays its readings on the home page. Clicking on the three dots displays the greenhouse specifications page. Clicking the "Add New Greenhouse" button will allow the user to add a new greenhouse

Greenhouse specification Pages

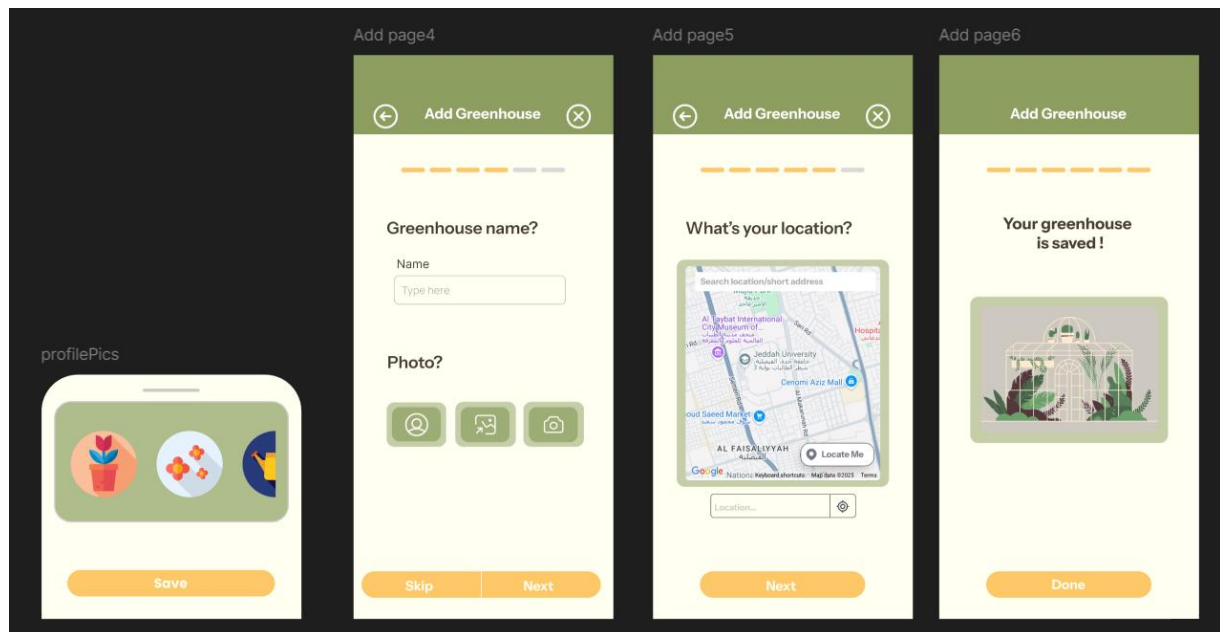


The greenhouse specification page displays all the information present related to a specific greenhouse. Moreover, it has addition information that the user can fill manually that were present when adding the greenhouse to make the process easier on the user.

Add Greenhouse Pages



The first step to adding a new greenhouse is to select the type of the green house. Following that the width, height, and length for the green house where the user is only allowed to enter numerical values with option to switch to the desired unit of measurement. After the user specifies in which direction the sun hits their green house with an interactive compass that points at the sun direction. Along with a progress bar at the top to keep the user informed on which step they're at.



Next, the user is prompted to enter a name for the greenhouse, with the option to skip this step. If skipped, a default name is assigned, which can be changed later. The user can also choose a picture from the icons displayed by the app, upload an image from their gallery, or take a new photo using the camera.

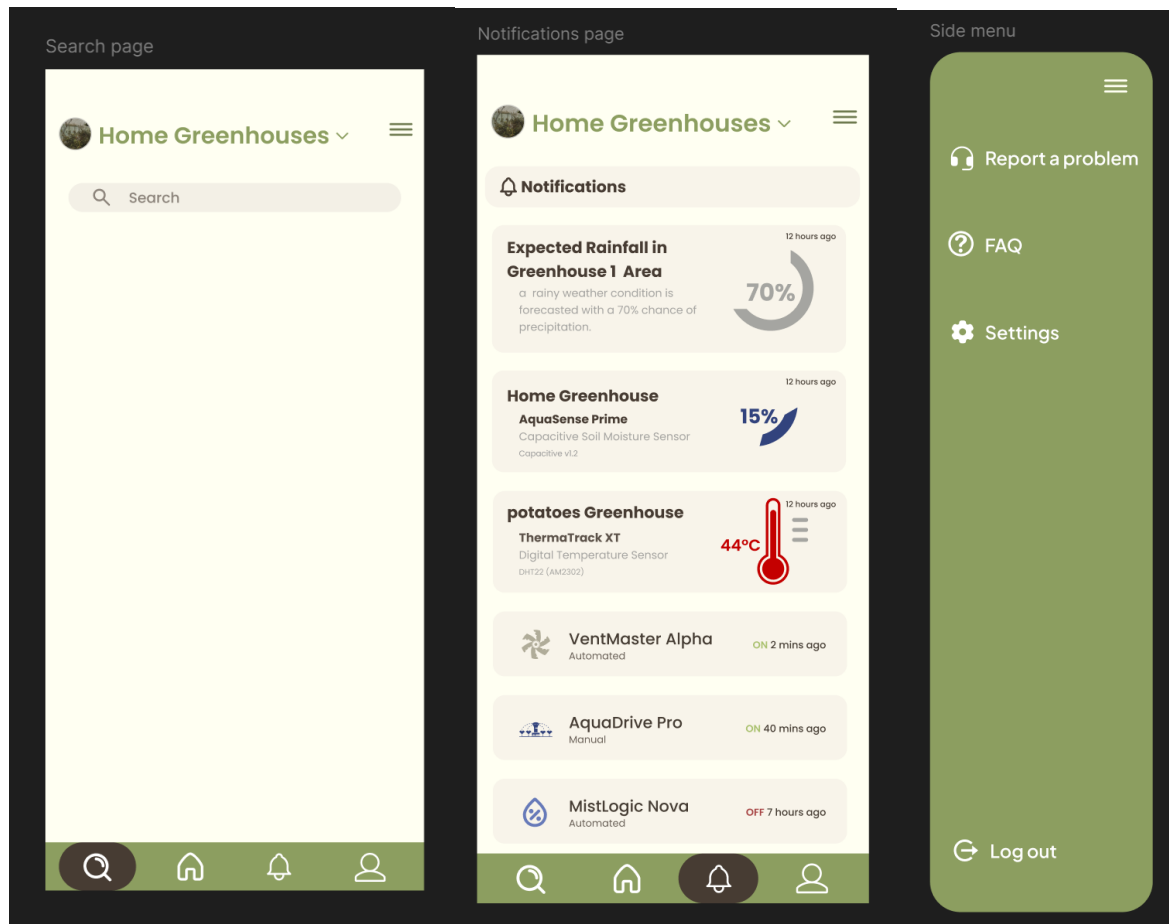
The user then selects the location of the greenhouse on an interactive map. They can either manually enter an address or use GPS to determine their current location.

Finally, a confirmation message appears indicating that the greenhouse has been saved. When the user presses the "Done" button, they are immediately returned to the main page of the app.

Additional Pages Added for Functional Completion

During the high-fidelity design process, several new pages were added that were not part of the initial low-fidelity prototype. These pages were necessary to fully support the system's core features, improve user experience, and ensure alignment with the defined business use cases.

The added interfaces are listed below:



Short Explanation of the Application/System

SeedSync is a smart mobile app that uses IoT to help users monitor and control greenhouse environments remotely. It displays real-time sensor data and allows users to manage devices like fans, pumps, and lights with ease. Key features include a motor control page for automation, a sensor readings page that prioritizes critical alerts, and a multi-greenhouse management system. Setup is made simple with visual guides, interactive animations. Users can view detailed specs, track system history, and search or edit greenhouse info, all through a user-friendly interface.

The system applies core HCI principles through its design features:

- ❖ **Constraints/ Error Prevention:** During greenhouse setup input fields for dimensions only accept numerical values, preventing users from entering invalid data.
- ❖ **Affordance:** The interactive sun-orientation compass and animated visuals clearly suggest their purpose and how users can interact with them.
- ❖ **Visibility:** On the sensor readings page, cards display brief summaries with arrows indicating they can be expanded, making the interaction immediately obvious.
- ❖ **Flexibility:** Users who don't wish to upload a photo or assign a custom name can skip those steps and use default icons and names provided by the system.

- ❖ **Inclusivity:** Users can choose between units (meters or feet) based on their preference or familiarity, making the app accessible to a wider audience from different regions.
- ❖ **Recognition Over Recall:** The use of familiar icons and a clean layout helps users understand the interface quickly without needing a tutorial.
- ❖ **Feedback:** Visual cues like progress bars, real-time updates, and confirmation messages let users know the system has received and processed their actions.
- ❖ **Prioritization of Information:** On the home page, sensor readings are sorted by urgency, so users can immediately focus on critical issues.
- ❖ **Design for Quick Recognition:** In the history page, automated actions are marked with a blue border, allowing users to distinguish them instantly.